

**RTG: electronic Computational Homotopy Theory Research
Community
Year 4 report**

FUNDED PARTICIPANTS

Funded participants in Year 4 were:

- Kiran Luecke, postdoctoral researcher. The project paid half of his academic year salary and also provided summer support.
- Jesse Keyes, graduate student. The project paid a portion of his financial support.

RESEARCH ACTIVITIES

Reading seminars. We hosted two research-oriented reading seminars. The topics were group actions on exotic spheres; and the secondary Steenrod algebra. The participants consisted of a mixture of graduate students, postdoctoral researchers, and tenure-track/tenured faculty. About half of the participants were located at U.S. institutions.

The organizers selected participants to give weekly talks on specific topics. We have had great success with this model of having participants take turns giving presentations.

Group actions on exotic spheres. The seminar was organized by J.D. Quigley and Oliver Wang (University of Virginia). The seminar attracted approximately 20 participants.

Secondary Steenrod algebra. The seminar was organized by Dan Isaksen (Wayne State University), Eva Belmont (Case Western Reserve University), and Martin Frankland (University of Regina). The seminar attracted approximately 30 participants.

Research seminar. We held a regular research seminar that met once per month for two talks. The seminar was managed by Guchuan Li (Peking) and J.D. Quigley (Virginia).

The 16 speakers were evenly split between postdoctoral researchers and tenure-track/tenured faculty. Two-thirds of the speakers were located in the United States, with additional speakers from China, Spain, and the United Kingdom.

Video recordings of most seminars are available to the public. Analytics suggest that the videos earn a significant on-going viewership.

Mini-courses. eCHT hosted a mini-course in May 2025, given by Ishan Levy, on the topic of K -theory and the telescope conjecture. Video recordings are available to the public. Analytics suggest that the videos earn a significant on-going viewership.

INSTRUCTIONAL ACTIVITIES

We offered two synchronous online courses in the 2025–2026 academic year. Each student was expected to participate fully in the course, just as if it were an in-person course at the student’s home institution. Each student earned a formal final grade at the end of the semester. These grades were reported directly to the students’ home institution advisors.

We advertised the courses through a variety of channels, including subject-specific mailing lists such as `algtop-1`.

Stable homotopy theory. Funded postdoctoral researcher Kiran Luecke taught a course on stable homotopy theory. It was intended for mid-career graduate students who were already familiar with graduate-level algebraic topology. Fifteen graduate students enrolled in the course, of which fourteen were located in the United States. Students took turns giving presentations on the topic of the week. The presentations were based on outlines provided by the instructor. Students also worked each week in small groups on problem sets.

The same course was previously offered in Winter 2024 and Winter 2025. This course is eCHT’s most successful instructional activity. In every iteration, the applicant pool was particularly strong. There is clearly demonstrated demand for a yearly offering on a permanent basis.

The course materials are available to the public on the eCHT website. The course is a valuable resource to widen access to stable homotopy theory. More generally, the course provides a viable model for courses on a variety of topics that are intended for mid-career graduate students. There is great demand for such courses from students because many institutions are unable to offer low-enrollment courses for advanced graduate students. Advanced graduate students are very mature students, and they take ownership for their own learning. This maturity overcomes many of the deficiencies of online instruction.

Kan seminar. Funded postdoctoral researcher Kiran Luecke led a Kan seminar in Fall 2025. Fourteen graduate students participated in the seminar. Each student gave a presentation about a classic paper in algebraic topology. Ten of the students were enrolled at institutions in the United States. The seminar is an electronic version of the well-known Kan seminar at MIT.

eCHT has offered online Kan seminars every year since 2019. They have been one of our most popular activities. They are well-suited for graduate students who are making the difficult transition from coursework to independent study. We plan to offer Kan seminars yearly in the future, provided that we can identify experienced researchers who are willing to provide leadership for the seminars.

ASSESSMENT

We conducted a formal evaluation for each course and reading seminar. Participants were invited to provide anonymous feedback. Response rates were appropriately high. Overall, participants were enthusiastic about the experiences. Some participants offered specific suggestions that we intend to act upon in future years.

BUDGET AND ADMINISTRATION

The project administration requires cooperation amongst a variety of departments at different institutions, as well as their sponsored programs offices. There was some concern that these complicated interactions could lead to ineffective management. This turned out not to be an issue. Some effort was involved to make sure that all units communicated with each other, but there were no major problems of this nature.

CONDUCT

Unfortunately, many online communities degenerate into unhealthy, negative interactions. We are consciously building a healthy online community. This requires the leaders to take active roles in setting constructive tones. We are pleased with the results of our efforts.

We wrote and posted an official eCHT Code of Conduct:
s.wayne.edu/echt/echt-code-of-conduct.

We offer an online reporting tool, with optional anonymity.

In the 2025–2026 academic year, no incidents were reported.